

Data Science

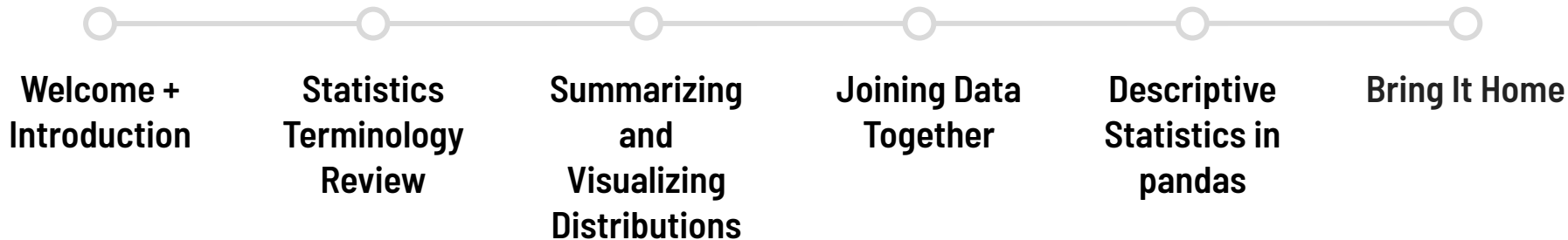
Exploring and Joining Data with pandas

Exploratory Data Analysis



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Understand basic statistical vocabulary and define different types of data.

2

Compare measures of central tendency and spread.

3

Summarize and visualize the shape of a variable.

4

Explain different kinds of joins, and merge data sets together using **pandas**.

5

Explore descriptive statistics of a data set using **pandas**.

Statistics Terminology Review

Statistics Terminology

Statistic

A value that has been computed from data.

Descriptive Statistics

Summarizing one or more variables from a dataset.

Variable

A single **column** of data, e.g. customer age.

Observation

A single **row** of data, e.g. a customer record.

Terminology Visualized

Customer ID	Customer Forename	Customer Surname	Customer Age
505	John	Smith	55
196	Templeton Arthur	Peck	43
947	H. M.	Murdock	36
952	Bosco Albert	Baracus	31

Terminology Visualized

Observation	Customer ID	Customer Forename	Customer Surname	Customer Age
	505	John	Smith	55
	196	Templeton Arthur	Peck	43
	947	H. M.	Murdock	36
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Variable

Terminology Visualized

Customer ID	Customer Forename	Customer Surname	Customer Age
505	John	Smith	55
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Summary statistic

Mean age: **41.25**

What is the difference between:
a **discrete** and
a **continuous**
variable?

Data Types

Continuous types

Interval

Data on a continuous scale.
Zero does not mean an absence.
e.g. temperature ($0^{\circ}\text{C} \neq$ "no temp")

Ratio

Data on a continuous scale.
Zero means an absence.
e.g. height, weight

Discrete / Categorical types

Nominal

Discrete categories, with no ordering.
e.g. color, job title

Ordinal

Discrete, ordered categories.
e.g. Likert data (ratings out of 5)

Central Tendency and Spread

With a new data set, for each numeric column we are often interested in:

- **Central tendency** - what is a typical value for this column?
 - Mean
 - Median

- **Spread** - how much do the values in this column vary?
 - Standard deviation
 - Variance
 - Interquartile range

How do we measure the “average”?

Given the data:

$$\mathbf{x} = (x_1, x_2, \dots, x_n)$$

The (sample) **mean** is:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

How do we measure the “average”?

Given the data:

$$\mathbf{x} = (x_1, x_2, \dots, x_n)$$

The (sample) **median** involves:

1. Sorting n observations in ascending order
2. The median is:
 - a. Observation $(n + 1) / 2$ if n is **odd**
 - b. The mean of observation $n/2$ and $n/2 + 1$ if n is **even**

Mean vs Median

Customer ID	Customer Forename	Customer Surname	Customer Age
505	John	Smith	55
196	Templeton Arthur	Peck	43
947	H. M.	Murdock	36
952	Bosco Albert	Baracus	31

Which is right?

Mean age: **41.25**
Median age: **39.5**

Mean vs Median

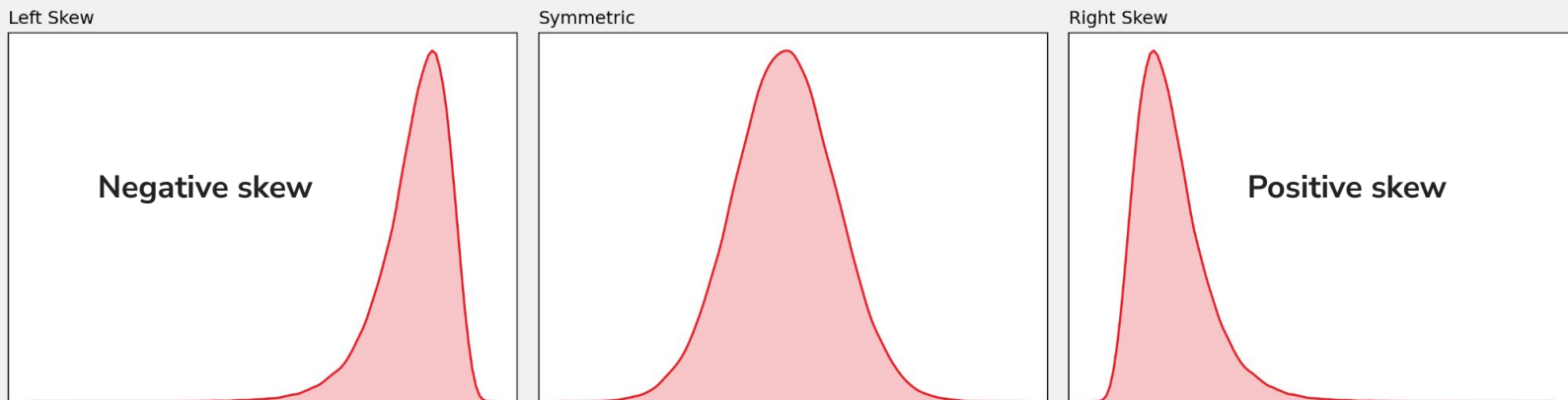
Customer ID	Customer Forename	Customer Surname	Customer Age
505	John	Smith	55
196	Templeton Arthur	Peck	43
947	H. M.	Murdock	36
952	Bosco Albert	Baracus	31
198	Yo	Da	899

New observation, what happens to the average?

Mean age: ~~41.25~~ 212.8
Median age: ~~39.5~~ 43

Skewness

If our mean and median are far apart, it is often because the data is **skewed**:



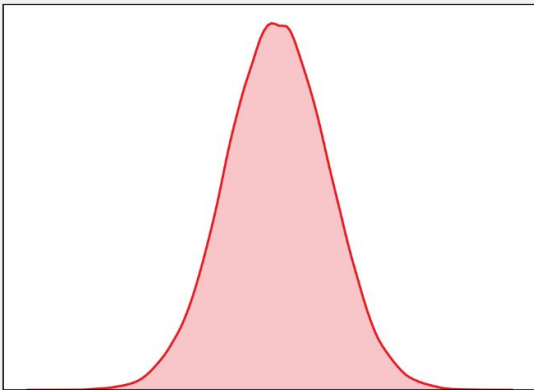
We want to know if our data is skewed so we can either transform it or use appropriate measures of central tendency (e.g. median instead of mean).

The Mode

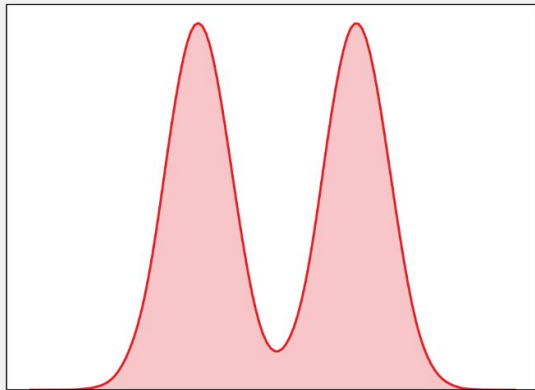
The mode is the most common value, which is useful:

1. To find the **most common category** in categorical data.
2. To identify if our continuous data is “multimodal”.

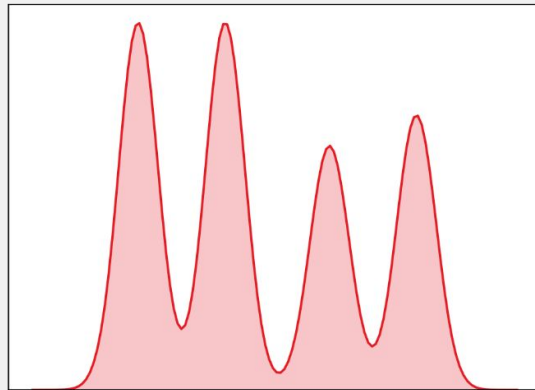
Unimodal



Bimodal



Multimodal



Variance

As well as knowing a central point to our data, we want to know how much observations **vary** (from a central point, e.g. the mean).

We probably want to know:

$$x_i - \bar{x}$$

This is a **deviation**.

Variance

The average deviation is therefore:

$$\frac{1}{n} \sum (x_i - \bar{x})$$

Except here, positive and negative deviations cancel each other out.

Variance

One solution is to **square** these differences:

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

This is the **sample variance**.

The units are in squared distances, which is not ideal and we can fix that.

Standard Deviation

This is the sample **standard deviation**:

$$s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$$

This helps us understand how spread out our data is.

It can also be used to identify outliers.

Summarizing and Visualizing Distributions

The “Five-Number Summary”

Often knowing just the mean and standard deviation isn't enough, we want to know the *shape* of a variable, not just a summary.

We usually also care about:

- The **minimum** value.
- The **maximum** value.
- The **range** (between the minimum and maximum values).

Like the mean, these are all very sensitive to outliers.

Percentiles

The p^{th} percentile is the number at which $p\%$ of the data lie below that number.

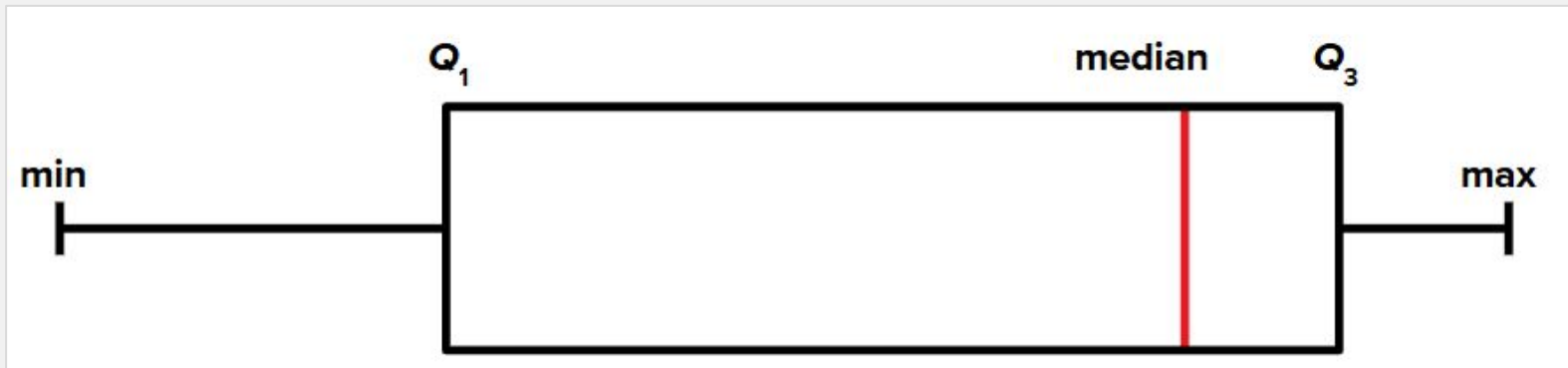
e.g. the 50th percentile is the **median**!

We often calculate the 25th and 75th percentiles (Q^1 and Q^3 respectively), and the difference between them, the **interquartile range** (IQR):

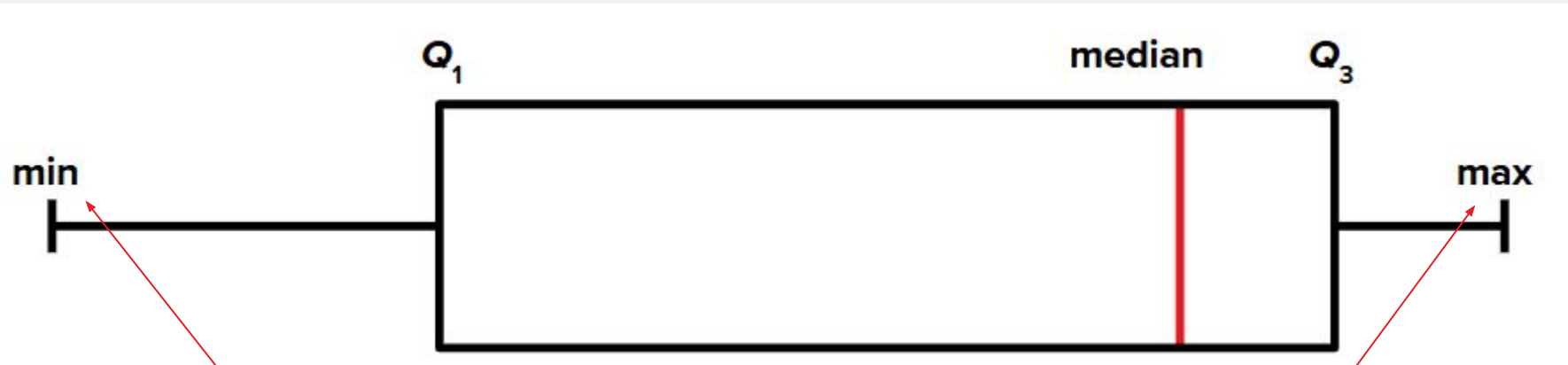
$$IQR = Q_3 - Q_1$$

The “five-number summary”

The combination of the minimum, Q_1 , median, Q_3 , and maximum is known as the **five-number summary** of a dataset. We can calculate the values or look at them graphically with a **box plot**.

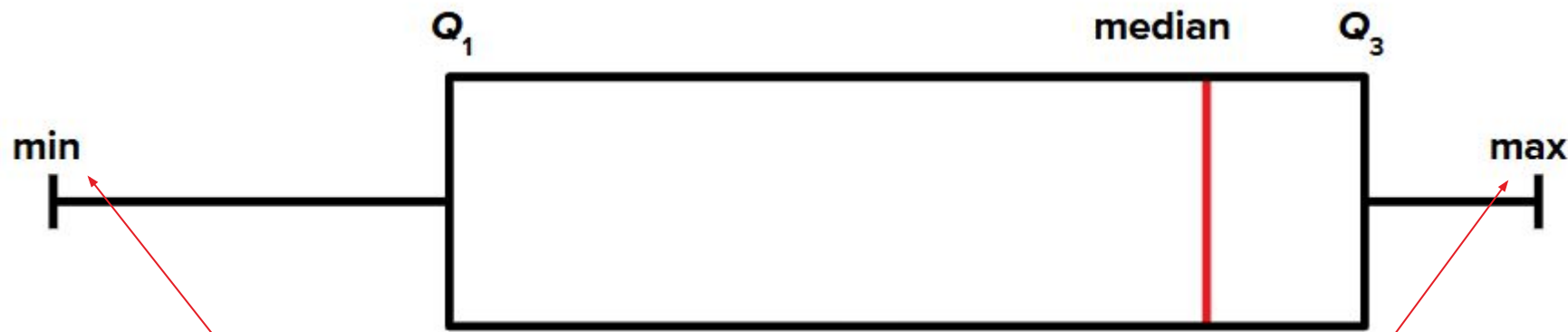


Box plots (continued)



Sometimes the “whiskers” are actually $Q_1 - 1.5 \cdot \text{IQR}$ and $Q_3 + 1.5 \cdot \text{IQR}$, so you can have points beyond them (to identify outliers)!

Box plots (continued)



Sometimes the “whiskers” are actually $Q_1 - 1.5 \cdot \text{IQR}$ and $Q_3 + 1.5 \cdot \text{IQR}$, so you can have points beyond them (to identify outliers)!

Histograms

An alternative to box plots to show the distribution of values.

Histograms can help us identify:

1

The general shape of the data (where are most of the values?).

2

Whether a variable is normally distributed (a bell curve).

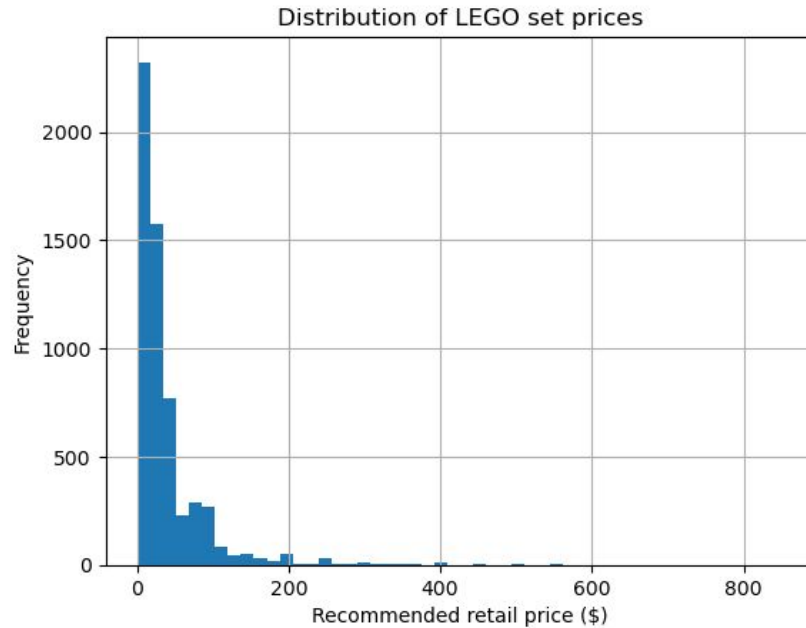
3

Whether a variable contains lots of outliers.

Histograms



How would we interpret this histogram?



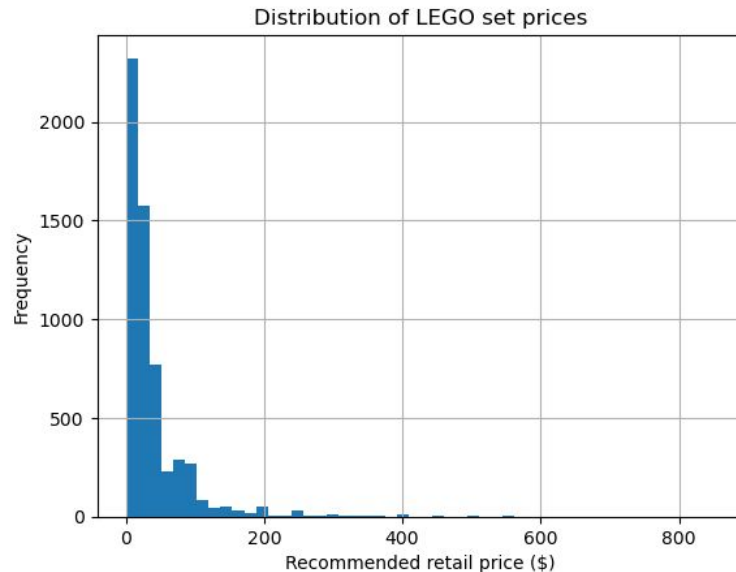
Histograms

How would we interpret this histogram?

- The most typical values in this dataset are near 0 (~less than \$20).
- The distribution has a **long tail**.
- Expensive sets (>\$200) exist but are rare.

Next steps:

- Investigate most expensive sets - are they outliers?
- Investigate 0-value sets - do they exist?



Descriptive Statistics

Let's see it for ourselves!

- Open **01-distributions.ipynb**



Joining Data Together

Joining data together

- Often we want to **combine** data sets together.
 - For example, enhancing a table with customer ID in it by adding customer information from another source.
- There are different **types of join**.
 - **Inner joins** only keep rows with data in both sources.
 - **Outer joins** (left/right join) keep rows regardless of a match in the other table.
- **pandas** allows joining data from **different sources**.

Joining data together

userId	movieId	rating	timestamp
1	1	5	1/1/22 1:00
2	1	4	1/1/22 2:00
2	2	2	1/2/22 11:00

INNER JOIN

userId	name	user_since
1	Tim Book	2/28/2019
2	Katie Sylvia	8/17/2020

INNER JOIN

movieId	title	year
1	Shrek	2001
2	The Hulk	2008

INNER JOIN

userId	movieId	rating	timestamp	name	user_since	title	year
1	1	5	1/1/22 1:00	Tim Book	2/28/2019	Shrek	2001
2	1	4	1/1/22 2:00	Katie Sylvia	8/17/2020	Shrek	2001
2	2	2	1/2/22 11:00	Katie Sylvia	8/17/2020	The Hulk	2008

Here we can use an **inner join** because every **userId** has a record in the user table.

Joining data together

userId	movieId	rating	timestamp
1	1	5	1/1/22 1:00
2	1	4	1/1/22 2:00
3	2	2	1/2/22 11:00

INNER JOIN

userId	name	user_since
1	Tim Book	2/28/2019
2	Katie Sylvia	8/17/2020

LEFT JOIN

movieId	title	year
1	Shrek	2001
2	The Hulk	2008

INNER JOIN

userId	movieId	rating	timestamp	name	user_since	title	year
1	1	5	1/1/22 1:00	Tim Book	2/28/2019	Shrek	2001
2	1	4	1/1/22 2:00	Katie Sylvia	8/17/2020	Shrek	2001
3	2	2	1/2/22 11:00	NULL	NULL	The Hulk	2008

In this scenario, user 3 does not exist in the users table.

Using a **left outer join** keeps all records, with **null** values for user 3's data.

Joining data in pandas

In pandas, once data is read in, it is a **DataFrame**.

DataFrames are the same regardless of source.

The original source of the DataFrame becomes irrelevant.

You can easily join one table from a CSV to another from a SQL database!

Joining data together in practice

- ❑ Determining what kind of join we need depends on the data
- ❑ The moment there is missing data involved, we need to consider outer joins (left/right)
- ❑ Do we want to keep **all data regardless of a match?**
 - Then we need a left/right join
- ❑ Is it more important to keep **only complete records?**
 - Then use an inner join

Descriptive Statistics in pandas

Descriptive Statistics in pandas

Let's see it for ourselves!

- Open **02-descriptive-statistics-in-pandas.ipynb**



Descriptive Statistics in pandas

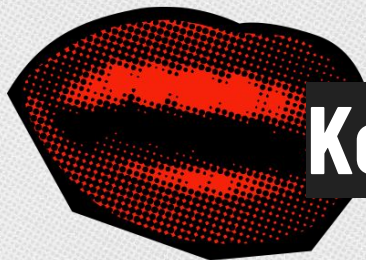


Now you try!

→ Open **03-exercise-descriptive-statistics-in-pandas.ipynb**



Bring It Home



Key takeaways

Descriptive Statistics in pandas

1

Terminology

Observations are usually rows.

Variables are usually columns.

A “statistic” is a summary measurement, like a mean or a median.

2

Summary statistics

Some statistics, like the mean, are influenced by outliers.

Some are more robust, like the median.

3

Distributions

Understanding the distribution of our data helps us get a broader picture.

Visualizing distributions lets us spot patterns and anomalies.

4

Descriptive statistics

pandas lets us calculate summary statistics.

pandas can also help visualize the distribution of our data.

Understanding summary statistics helps us answer business questions.

